The Alphaland Regional Government has six sectors which need fire protection. The response time is critical and adequate fire protection can be provided in each sector either by building a fire station in that sector, or by building a fire station in another sector which is no more than a 12-minute drive away (note that it is OK to have more than one sector with a fire station within 12-minute drive of any given sector). The time to drive between the centres of each pair of sectors "*i*" and "*j*" is given by t_{ij} (because of one-way streets and left-turns the times are not symmetric). The government also decided not to build a fire station if the population of a sector, denoted by P_i, is less than 50,000 inhabitants.

The cost of building a fire station is the same in each sector, and the government wishes to build the least number of fire stations making sure that each sector is served (by a fire station in that sector and/or in a different sector (or sectors) less than 12-minute drive away). Formulate an IP.

SOLUTION

We first define the following binary variable as in each sector either a fire station is built or not:

$$Y_i = \begin{cases} 1, & \text{if a fire station is built in sector } i \\ 0, & \text{otherwise} \end{cases}, \quad \text{for } i = 1, \dots, 6$$

Secondly, we define binary variables that show which sector is served by which other sector(s):

$$F_{ij} = \begin{cases} 1, & \text{if sector } i \text{ has a station serving sector } j \\ 0, & \text{otherwise} \end{cases}, \quad \text{for } i = 1, \dots, 6 \text{ and } j = 1, \dots, 6.$$

Then, the formulation will be as follows:

$$\text{Min} \quad z = \sum_{i=1}^{6} Y_i$$

$$\text{s.t.} \quad \sum_{i=1}^{6} F_{ij} \geq 1, \quad \forall j = 1, \dots, 6$$

$$F_{ij} - Y_i \leq 0, \quad \forall i, j = 1, \dots, 6$$

$$t_{ij} \cdot F_{ij} \leq 12, \quad \forall i, j = 1, \dots, 6$$

$$(P_i - 50{,}000) \cdot Y_i \geq 0, \quad \forall i = 1, \dots, 6$$

$$Y_i, F_{ij} \text{ binary } \forall i, j = 1, \dots, 6$$